
Position: Multiple Definitions & Unrealistic Assumptions of Model Collapse Distract from Real World Threats

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Abstract

The proliferation of AI-generated content online has fueled concerns over *model collapse*, a degradation in future generative models’ performance when trained on synthetic data generated by earlier models. Industry leaders, premier research journals and popular science publications alike have prophesied catastrophic societal consequences stemming from model collapse. In this position piece, we contend this widespread narrative fundamentally misunderstands the scientific evidence. We highlight that research on model collapse actually encompasses eight distinct and at times conflicting definitions of model collapse, and argue that inconsistent terminology within and between papers has hindered building a comprehensive understanding of model collapse. To assess how significantly different interpretations of model collapse threaten future generative models, we posit what we believe are realistic conditions for studying model collapse and then conduct a rigorous assessment of the literature’s methodologies through this lens. While we leave room for reasonable disagreement, our analysis of research studies, weighted by how faithfully each study matches real-world conditions, leads us to conclude that certain predictions of model collapse rely on assumptions that poorly match real-world conditions, and in fact several prominent collapse scenarios are readily avoidable. Altogether, this position paper argues that model collapse has been warped from a nuanced multifaceted consideration into an oversimplified threat, and that the evidence suggests specific harms more likely under society’s current trajectory have received disproportionately less attention.

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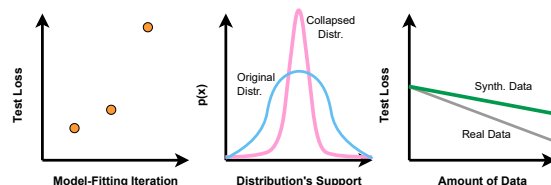


Figure 1. Model Collapse Has Been Defined in Multiple and Sometimes Conflicting Ways. By hand-annotating 28 prior research publications, we identify 8 definitions of model collapse (Sec. 2). The 8 definitions can be loosely grouped into three families: (1) the behavior of the test loss on real data over model-fitting iterations (left), (2) the deformation of the real data distribution over model-fitting iterations (center) and (3) the scaling behavior of the test loss with respect to typical scaling quantities such as the amount of data (right). By grounding these definitions in realistic conditions, such as data accumulation and increasing compute, we shift the focus from theoretical catastrophe toward identifying and mitigating the tangible threats of diversity and tail-data loss.

1. Introduction

The rapid surge of AI-generated content has sparked intense debate about potential ramifications of training future generative AI models on datasets containing synthetic data generated by previous models. One especially concerning prediction is *model collapse*, a phenomenon whereby future generative models fail due to being trained on synthetic data. Model collapse has captured attention at the highest levels of academia and industry: *Nature* prominently featured model collapse (Gibney, 2024) alongside accompanying research stating that AI models trained on synthetic data would suffer catastrophic degradation in performance (Shumailov et al., 2024), while prominent science and news outlets like *Scientific American* and the *Wall Street Journal* amplified these concerns, writing “a training diet of AI-generated text, even in small quantities, eventually becomes poisonous to the model being trained.” (Rao, 2023) and that “feeding a model text that is itself generated by AI is considered the computer-science version of inbreeding” (Seetharaman, 2024). Some industry leaders have highlighted model collapse as a critical challenge for society (Wang, 2024).

In this position piece, we argue this widespread narrative of a bleak model-collapsed future, filled with polluted pretraining data and useless generative models, oversim-

plifies or misinterprets the precise scientific claims and their underlying assumptions and mechanisms. Through careful analysis, we identify three critical gaps between the prevailing discourse and research reality:

First, we reveal that the term “model collapse” encompasses eight definitions of performance degradation in model-data feedback loops. This multiplicity of definitions, used inconsistently between papers and at times inconsistently within papers, has resulted in papers talking past one another, thereby hindering development of a comprehensive understanding of likely futures for frontier deep generative models. We argue specific failure modes should be explicitly identified and discussed alongside comparable results.

Second, we posit trends that we believe faithfully describe common practices of leading AI labs pretraining frontier AI systems on web-scale data: increasing compute, improving data quality, and expanding datasets of real and synthetic data. One key point that we emphasize here is that many prominent model collapse papers assume data are entirely deleted after each model-fitting iteration and that subsequent models are trained entirely on synthetic data generated by their predecessors, which we argue is not realistic.

Third, we weigh different results in the model collapse literature based on the plausibility of their assumptions along multiple axes of consideration. We argue that some definitions of model collapse do not correspond to catastrophic outcomes under our realistic assumptions, and most concerning collapse predictions emerge from implausible experimental setups. However, there are very real threats to tails of the data distribution that should be taken seriously.

We argue that model collapse has been inflated from a precise and important technical consideration into a mischaracterized threat. While synthetic data poses genuine challenges that warrant careful study, our analysis reveals that the most widely stated collapse scenarios can be avoided through standard ongoing practices in model development and dataset curation. Instead of worrying about unrealistic catastrophic notions of model collapse, by adopting a more realistic perspective, we can re-orient to focus on the real issues of diversity collapse happening now (Zhang et al., 2024; Padmakumar & He, 2024; Murthy et al., 2024; Wu et al., 2024). This position paper aims to clarify the scientific discourse around model collapse, propose best practices for future work on the subject, and redirect research attention towards mitigating failure modes.

2. Definitions of Model Collapse

We begin with a non-obvious but critical point: **the model collapse literature has at least eight different definitions.** As evidence, we hand-annotated twenty-eight prominent prior research publications on model collapse to determine

which papers offer explicit definition(s) of model collapse (Fig. 2), where we define explicit as either (1) any mathematical definition, or (2) any precise verbal description of the failure behavior *independent* from results. We additionally categorize which definition(s) of model collapse each paper uses, perhaps implicitly, across any and all mathematical and empirical results (Fig. 2 bottom). We find that many papers do not offer explicit definitions of model collapse and also sometimes use multiple definitions, leading to a lack of specificity and apparent contradictions both within single works and across multiple works.

Before delving into the eight definitions, we first define some shared terminology. In general, we consider model-data feedback loops whereby f_t is the t -th generative model, and we study the behavior of a sequence of generative models $(f_t)_t$ that are iteratively fit to data and then sampled from. We call data sampled from a generative model *synthetic data*. We can evaluate the quality of a generative model in multiple ways. One prominent way is via the *population risk*, defined as the expected loss over the entire real data distribution $\mathbb{E}_{x \sim P}[\ell(f_t(x))]$, where x is some real datum, P is the real data distribution, and ℓ is the loss function (Vapnik, 1991). Another way to evaluate the quality of a generative model is by its *tail risk*, defined as the expected loss conditioned on tail events $\mathbb{E}_{x \in \text{Tail}(P)}[\ell(f_t(x))]$, where $\text{Tail}(P)$ informally represents real data with low probability. Population risk provides a holistic view of performance, but may mask specific failure modes (Xu, 2024; Kozerawski et al., 2022), whereas tail risk can reveal degradations in edge cases even when population risk remains stable (Hoffmann & Börner, 2020; Xu, 2024). There are many other salient properties of generative models one can use to assess whether the models collapse.

1. **Catastrophic Increase of Population Risk** (Dohmatob et al., 2025; Bertrand et al., 2024; Kazdan et al., 2024b): Perhaps the most colloquial definition, model collapse is a critical and rapid degradation in model performance due to the presence of synthetic data, as measured by population risk. We note that what constitutes catastrophic is often undefined.
2. **Any Increase of Population Risk** (Alemohammad et al., 2024; Dohmatob et al., 2025): Under this strict definition, model collapse occurs if there is *any* increase in population risk when training with synthetic data compared to training with real data alone.
3. **Asymptotically Diverging Population Risk** (Gerstgrasser et al., 2024; Kazdan et al., 2024b; Dey & Donoho, 2024): This definition considers model collapse to occur when the population risk grows without bound over successive model-fitting iterations. This represents a fundamentally unstable learning dynamic

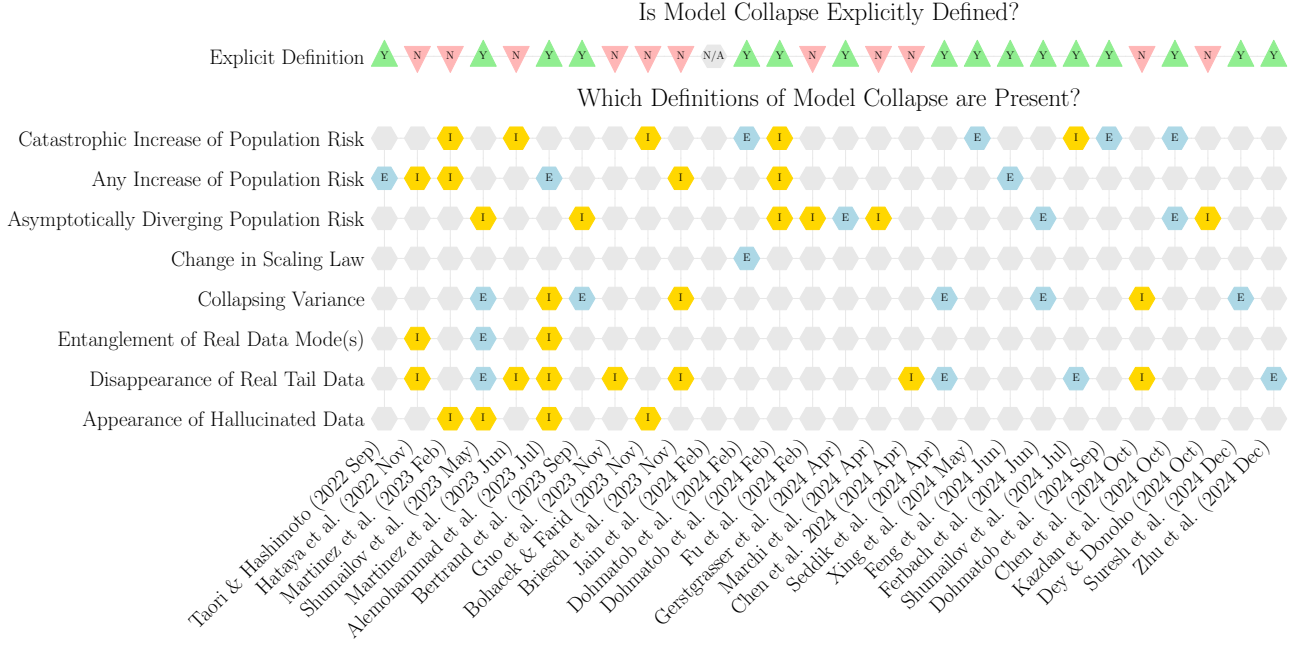


Figure 2. Model Collapse Has Been Defined in Multiple and Sometimes Conflicting Ways. We conduct a meta-analysis of research papers on model collapse. Top: We identify which papers offer *any* explicit definition of model collapse (**Yes (Y)** or **No (N)**). Bottom: We identify which definition(s) of model collapse each paper uses for its empirical and/or mathematical results, either **explicitly (E)** or **implicitly (I)**. Our annotations reveal that model collapse research is based on multiple definitions that we will show sometimes conflict between papers and even within individual papers.

where each iteration of synthetic data generation and training leads to progressively worse performance.

4. **Collapsing Variance** (Alemohammad et al., 2024; Shumailov et al., 2023; Bertrand et al., 2024) Model collapse here is when variance (or diversity) trends towards 0 and the learned distributions tend towards delta-like functions over model-fitting iterations.
5. **Change in Scaling Law** (Dohmatob et al., 2024b): Model collapse occurs if the governing scaling behavior changes due to the presence of synthetic data. Specifically, model collapse occurs if the relationship between model performance and training data size deviates from the expected scaling behavior observed with real data.
6. **Disappearance of or Entanglement of Real Data Mode(s)** (Alemohammad et al., 2024): Sometimes called “Mode Collapse” (Goodfellow et al., 2014; Lucic et al., 2018; Brock et al., 2019), model collapse here is defined by the presence of synthetic data preventing the model from learning particular modes of the real data distribution or causing the model to blur different data modes together.
7. **Disappearance of Real Tail Data** (Shumailov et al., 2023; Wyllie et al., 2024; Shumailov et al., 2024): Sometimes called “coverage collapse” (Zhu et al.,

2024), model collapse here occurs when synthetic data leads to the under-representation of data from the tail of the distribution, leading to models that can only handle common cases but fail on rare ones. The disappearance of real tail data can be more subtle and more narrow than the generative model losing all diversity (Def. 4). This definition presupposes that the bulk of the distribution stays roughly in place while low-probability mass is selectively eroded, distinguishing it from a wholesale shift in which bulk and tails move together.

8. **Appearance of Hallucinated Data** (Shumailov et al., 2023; Alemohammad et al., 2024; Bohacek & Farid, 2023): Model collapse occurs when the sequence of models begin producing fully-synthetic data not supported by the original real data’s distribution.

We note that the definitions can themselves be loosely clustered into three families (Fig. 1): (i) population risk degrading (Definitions 1, 2 and 3), (ii) distributions deforming from their original shape (Definitions 4, 6, 7, and 8), and (iii) decreasing value from additional data (Definition 5).

2.1. Intra-Paper Definitions Can Cause Confusion

The differences between different definitions of model collapse can be slippery, and understandably, authors sometimes move between them in the course of a paper. In this

section, we use a prominent prior work to demonstrate how easy it can be to slip between definitions. Our intention is not to call out this specific work, but rather demonstrate how a seemingly reasonable treatment of model collapse definitions can have implications for interpreting results.

Shumailov et al. (2023) admirably provides explicit definitions of two different types of model collapse: “We separate two special cases: early model collapse and late model collapse. In *early model collapse* the model begins losing information about the tails of the distribution; in the *late model collapse* the model entangles different modes of the original distributions and converges to a distribution that carries little resemblance to the original one.” These correspond to our Defn. 7 and 6, respectively. However, the paper presents results on model collapse that fall under different definitions. Firstly, Shumailov et al. (2023) demonstrate Definition 3 in their Sections 4.2 and 4.3. We lightly generalize their results for clarity and generality. We consider repeatedly fitting multivariate Gaussians to data and sampling from the fitted Gaussians. We begin with n real data drawn from a multivariate Gaussian with mean $\mu^{(0)}$ and covariance $\Sigma^{(0)}$:

$$X_1^{(0)}, \dots, X_n^{(0)} \sim_{i.i.d.} \mathcal{N}(\mu^{(0)}, \Sigma^{(0)}).$$

For model fitting, we compute the unbiased mean and covariance of the most recent data:

$$\begin{aligned} \hat{\mu}^{(t+1)} &\stackrel{\text{def}}{=} \frac{1}{n} \sum_{j=1}^n X_j^{(t)} \\ \hat{\Sigma}^{(t+1)} &\stackrel{\text{def}}{=} \frac{1}{n-1} \sum_{j=1}^n (X_j^{(t)} - \hat{\mu}^{(t+1)})(X_j^{(t)} - \hat{\mu}^{(t+1)})^T \end{aligned}$$

and then draw n new synthetic samples from a Gaussian with the most recently fit parameters. In this model-data feedback loop, Shumailov et al. (2023) proved that as the model-fitting iteration $t \rightarrow \infty$, the population risk as measured by the expected squared Wasserstein distance between the most recent multivariate Gaussian and the original multivariate Gaussian diverges asymptotically:

$$\mathbb{E}[\mathbb{W}_2^2(\mathcal{N}(\hat{\mu}^{(t)}, \hat{\Sigma}^{(t)}), \mathcal{N}(\mu^{(0)}, \Sigma^{(0)}))] \rightarrow \infty.$$

This result demonstrates that the population risk diverges asymptotically, which aligns with neither of the two definitions of model collapse stated at the outset of the paper. Indeed, the population risk diverges due to a third definition of collapse. Recall that the squared Wasserstein distance between two Gaussians has two terms: a contribution from the means and a contribution from the covariances.

$$\begin{aligned} \mathbb{E}[\mathbb{W}_2^2(\mathcal{N}(\hat{\mu}^{(t)}, \hat{\Sigma}^{(t)}), \mathcal{N}(\mu^{(0)}, \Sigma^{(0)}))] &= \underbrace{\|\hat{\mu}^{(t)} - \mu^{(0)}\|_2^2}_{\rightarrow \infty} \\ &+ \underbrace{\text{Tr}(\hat{\Sigma}^{(t)} + \Sigma^{(0)} - 2((\Sigma^{(0)})^{1/2} \hat{\Sigma}^{(t)} (\Sigma^{(0)})^{1/2})^{1/2})}_{\rightarrow 0}. \end{aligned}$$

Thus, the population risk diverges because the sequence of means $(\hat{\mu}^{(t)})_t$ randomly walks away from the ground truth mean $\mu^{(0)}$, while the covariance term vanishes; the entire distribution rigidly translates rather than changing shape. This is one example in which a casual reader might fail to notice that the asymptotic divergence is driven by *mean drift*, a mechanism distinct from the two failure modes Shumailov et al. (2023) define: *early* model collapse, the selective loss of tail information while the bulk of the distribution stays in place, and *late* model collapse, the merging of multiple modes into fewer. Because the Gaussian has a single mode throughout, no tails are selectively lost and no modes merge; although one might loosely call the drifting mean “the mode moving away,” this conflates rigid translation of the whole distribution with the selective tail or mode loss those definitions describe.

The same paper also demonstrates how definitions of model collapse can go beyond subtle confusion to explicit contradiction. Shumailov et al. (2023) experimentally demonstrate model collapse under a *fourth* definition of model collapse: in their Section 5.2, the authors consider fine-tuning sequences of language models on the models’ own generations. The authors’ Figure 10 shows that while the population risk initially increases, it then decreases and converges to a plateau about 12.5% – 50% above the population risk of the first model. The authors instead find that “Over the generations, models tend to produce samples that the original model trained with real data is more likely to produce.” We believe that under Definitions 3, 6, and 7, these results suggest that model collapse has not occurred. However, the authors label this result as model collapse because later generations begin introducing fully synthetic tail data over time: “later generations start producing samples that would never be produced by the original model, i.e., they start misperceiving reality based on errors introduced by their ancestors.” This appearance of hallucinated data qualifies as model collapse under Defn. 8.

When a paper shifts definitions without explicitly acknowledging the change, it creates a cascade of problems undermining scientific clarity. Readers interpret results through the lens of the initially stated definitions, creating a false sense that all discussed phenomena represent the same underlying issue when they may be fundamentally distinct. This leads to misattribution of causes and effects, as demonstrated in the Gaussian example, where the asymptotic divergence of the population risk stems from mean drift, the rigid translation of the entire distribution, rather than from the selective tail or mode loss that the term “model collapse” is often taken to imply. Such definitional inconsistency fosters overgeneralization of results, hampering cross-study comparisons and potentially prompting inappropriate technical responses or policy decisions. Most critically, when technical concepts like model collapse require precise charac-

terization, unstated definitional shifts prevent the formation of a stable framework for interpreting claims, ultimately contributing to broader confusion about which phenomena deserve concern and how they might be addressed.

2.2. Inter-Paper Definitions Can Cause Confusion

To demonstrate how different researchers can look at the same results and reach different conclusions regarding model collapse, Alemohammad et al. (2024) studied a FFHQ-StyleGAN2 (Karras et al., 2020) trained on synthetic data and found that the population risk as measured by Frechet Inception Distance (FID) (Heusel et al., 2018) increased $2\times$ by the 5th model-fitting iteration and then plateaued. The authors declared this result constituted model collapse because the authors had implicitly defined model collapse as *any* increase in the population risk (Definition 2). Gerstgrasser et al. (2024) then questioned this claim that the models had collapsed, writing, “Figure 7 from Alemohammad et al. (2024) shows that linearly accumulating data (“Synthetic augmentation loop”) causes poor behavior to plateau with the number of model-fitting iterations [...] We believe is that our evidence and their evidence is more consistent with the conclusion that accumulating data avoids model collapse and does not merely delay it.” This apparent disagreement was because Gerstgrasser et al. (2024) had defined model collapse as asymptotically diverging population risk (Definition 3). Thus, while looking at the exact same figure, the researchers came to differing conclusions because they were operating under different definitions.

2.3. Stating and Adhering to Definitions Improves Clarity and Drives Progress

In this position paper, our intention is not to argue in favor of specific definitions of model collapse or call out authors whose definitions we disagree with. Rather, we hope to emphasize that model collapse is a multifaceted phenomenon: under the same results, model collapse can simultaneously “occur” and “not occur” in different researchers’ opinions; in Appendix A, we include a case study of how confusing and entangled scientific insights can become on account of different definitions and methodologies. This makes building a comprehensive understanding of model collapse difficult, which can be especially concerning to specific communities. For instance, search providers like Google, Bing, and Perplexity may pay an especially high penalty if their models are trained on hallucinated facts, whereas the disappearance of real tail data can disproportionately affect marginalized groups or historically disadvantaged communities (Blodgett et al., 2016; Bender & Friedman, 2018; Noble, 2018; Shah et al., 2020; Jo & Gebru, 2020; Koenecke et al., 2020; Bender et al., 2021; Hutchinson et al., 2021; Ji et al., 2023). By clarifying different definitions of model collapse and adhering to those definitions, researchers can build a better

understanding of model collapse with greater nuance for what causes different failures modes and what actions can be taken to prevent each. Together, these definitions can form a “model collapse profile” which can characterize the different ways in which models worsen over time. Research on the harms of synthetic data should explicitly note the relevant aspects of the model collapse profile they study.

3. Realistic Conditions for Studying Model Collapse

Our goal is to understand likely outcomes as humanity pre-trains future frontier AI systems on web-scale datasets containing a mixture of real and synthetic data. We focus on pre-training both because the literature has (Shumailov et al. (2023)’s “What will happen to GPT-n once LLMs contribute much of the language found online?”) and because pre-training’s tremendous capital and operating expenses render missteps extremely costly. Motivated by this goal, we posit trends that we believe describe the current trajectory of pre-training practices of frontier AI systems:

1. **Increasing Pre-training Compute:** The total floating point operations used for pre-training has been rapidly increasing. For example, Meta pre-trained Llama 1 using 2k GPUs, Llama 2 using 4k GPUs and Llama 3 using 16k GPUs (Goyal, 2024), and OpenAI recently announced a \$500B initiative to increase compute capacity of the United States (OpenAI & SoftBank, 2025), a fraction of which will be allocated for pre-training.
2. **Increasing Pre-training Data:** The amount of data used has been rapidly increasing. For example, Meta’s series of Llama language models were pre-trained on increasing amounts of data: 1.4T tokens for Llama 1, 2T tokens for Llama 2, and most recently, 15T tokens for Llama 3. While pre-training data will inevitably max out, recent estimates place the total number of available language pre-training tokens at 1 quadrillion (1000 trillion) tokens (Villalobos et al., 2024).
3. **Increasing Quality of Pre-training Data:** Over time, pre-training data is becoming higher quality on account of pre-training data teams developing better filtering techniques to ensure high-quality training data (Gao et al., 2020; Penedo et al., 2024; Li et al., 2024). Moreover, whatever synthetic data is shared online is increasingly higher quality since models are improving over time (Kiela et al., 2021; 2023; Maslej et al., 2024).
4. **Synthetic Data Accumulating Alongside Real Data:** Real data are not deleted en masse after each model-fitting iteration. When synthetic data are generated, they amass alongside prior data and new real data.

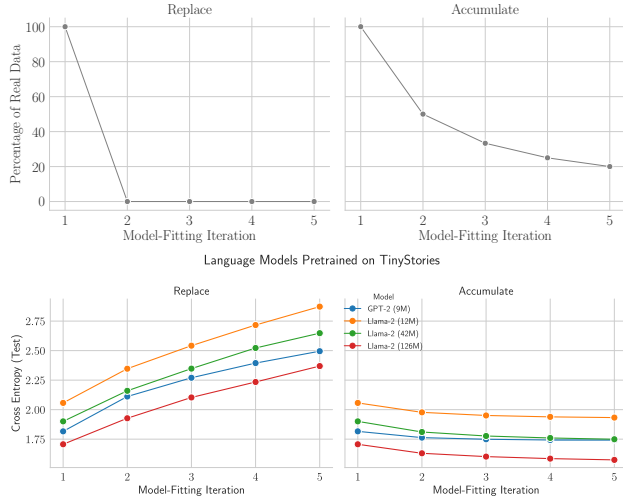


Figure 3. Dimensions of Consideration for Model-Data Feedback Loops: Propagation of Data Over Time and Proportion of Real Data Over Time. When data are *replaced* after each model-fitting iteration (left), the proportion of real data immediately becomes zero after the first iteration, whereas when data instead *accumulate* (right), the proportion of real data falls asymptotically to zero. Gerstgrasser et al. (2024); Kazdan et al. (2024b); Dey & Donoho (2024) showed that replacing data over time causes the population risk to diverge, whereas accumulating data avoids diverging population risk. In these works, synthetic data are assumed to grow linearly over time, contributing n samples per model-sampling iteration. Credit: The bottom figure is copied from Gerstgrasser et al. (2024) with permission.

5. **Decreasing Proportion of Real Data:** The fraction of real data relative to total (real plus synthetic) data is decreasing over time. However, whether the fraction of real data will asymptote to zero is unclear, a point we will return to later.

We believe researchers and policy makers interested in societal implications of model collapse should focus on research that adhere to these conditions as faithfully as possible.

4. Key Dimensions of Consideration for Model-Data Feedback Loops

Many of the studies we assess use simplified settings such as Gaussians or discrete distributions. We treat these as sources of mechanistic insight and paradigm contrasts rather than direct predictions for frontier systems: what transfers to real-world practice is the *conditions* under which collapse occurs (severe under full data replacement with fixed compute, absent or mild under data accumulation with growing compute), not the specific numbers from a simplified setting.

4.1. Propagation of Data Over Time

Early work that sounded the alarm about model collapse (Martínez et al., 2023; Alemohammad et al., 2024; Bohacek & Farid, 2023; Shumailov et al., 2023; Briesch et al., 2023; Bertrand et al., 2024) assumed that data propagate in a particular way: after training a model, all existing data are deleted, new data are sampled from the new model, and the next model is trained solely on this fresh synthetic data. Subsequent authors called this the *replace* paradigm (Gerstgrasser et al., 2024; Kazdan et al., 2024b; Dey & Donoho, 2024) because data are entirely replaced after each model-fitting iteration. When data are replaced, researchers demonstrated multiple harmful outcomes: variances collapse, real data tails disappear, population risk diverges, etc. (Fig. 3 left). This particular assumption of how data propagate over time is highly unrealistic: **After a model finishes training, the entire internet is not deleted, nor is the next model necessarily trained solely on its predecessor’s outputs.** Rather, a more realistic assumption is that synthetic data from each model *accumulates* on the internet alongside real data and past synthetic data such that all can be used for training the next model. To some, these differences might seem insignificant, but each produces vastly different asymptotic behavior in terms of population risk: Gerstgrasser et al. (2024) showed empirically and Kazdan et al. (2024b) and Dey & Donoho (2024) showed mathematically that population risk diverge if data are replaced, but population risk does not diverge if data instead accumulate (Fig. 3 right).

A slightly different but perhaps more realistic data propagation assumption is that real and synthetic data accumulate, but future models are trained on a downsampled proportion of the total available data (Kazdan et al., 2024b). This *accumulate-subsample* paradigm represents a middle ground between replace and accumulate: while test loss appears to stabilize, no analytic theory has been proven in this case to date. Individual beliefs about which data paradigm is most reflective of reality can influence perceptions of how model collapse will unfold in the future. However, to our knowledge, non-population risk-based notions of model collapse have not been connected to assumptions about how data propagate, leaving an important question open.

4.2. Proportion of Real Data Over Time

ChatGPT alone produces 1/1000 of all words produced by humanity each day (Altman, 2024); as these models proliferate, over time, future generative models could produce vastly more data than humanity for training future models. Thus, the proportion of future real data plays a crucial role in the debate over the effects of model collapse. Bertrand et al. (2024) claimed that the population risk will not asymptotically diverge so long as the proportion of real data remains lower-bounded above 0 (in addition to other

conditions). In Dohmatob et al. (2025)’s view, *any* synthetic data causes “a critical degradation” to future models, writing model collapse “generally persists even when mixing real and synthetic data, as long as the fraction of training data which is synthetic does not vanish” and that “model collapse cannot generally be mitigated by simple adjustments [...] unless these strategies asymptotically remove all but a vanishing proportion of synthetic data from the training process.”

Results like these draw attention to what proportion of real data is necessary to avoid collapse, which is a valid consideration. However, correctly interpreting these results takes care. For instance, Bertrand et al. (2024)’s condition is *sufficient*, not necessary; it should *not* be understood as saying that model collapse is inevitable unless real data remains a non-zero proportion. Moreover, contrary work by Gerstgrasser et al. (2024), Marchi et al. (2024), and Kazdan et al. (2024b) established conceptually stronger guarantees: when data *accumulate*, but human data asymptotically occupy a vanishing fraction of the internet, the population risk will likely not diverge (in some cases, with an additional requirement that the rate of AI data generation does not grow super-linearly) (Fig. 3). Relatedly, Gillman et al. (2024) also show that the proportion of real data can asymptotically approach zero using a function to “correct” synthetic data towards the real data distribution. Due to uncertainty about the rate of synthetic data generation, one potential solution is to sequester real training data for future use, when the internet still contains an abundance of human-generated data; however, frontier AI labs have already collected such data, meaning no additional work is required.

4.3. Model Training Assumptions

While multiple works claim that models do not collapse under certain settings or propose interventions to avoid collapse, we urge the explicit declaration of strong technical assumptions that are frequently unrealistic. As an example, Bertrand et al. (2024) and Gillman et al. (2024) study *iterative retraining*, where each model is initialized from its predecessor’s parameters and optimizer state. Under an additional assumption that the first model is sufficiently high performing, since each model is assumed to be close to its predecessor, collapse can be avoided. However, to the best of our knowledge, iterative retraining has not been used for any frontier AI model, including OpenAI’s GPT-2 (Radford et al., 2019), GPT-3 (Brown et al., 2020), GPT-4 (Achiam et al., 2023), Anthropic’s Claude 1, 2 or 3, Google’s PaLM 1 (Chowdhery et al., 2022), PALM 2 (Anil et al., 2023) or Gemini (Team et al., 2024), DeepSeek’s V3 (DeepSeek-AI et al., 2024). Thus, these sufficiency conditions for avoiding collapse are far from current practices.

For another example, Zhu et al. (2024) propose data edit-

ing as a collapse mitigation strategy and analyze a self-consuming linear model. Each model iteration fits $\hat{w}_n = X^\dagger \hat{Y}_n$, where X are fixed Gaussian covariates, and generates synthetic data $\hat{Y}_{n+1} = X \hat{w}_n + E_{n+1}$ where E_{n+1} are Gaussian errors and $\hat{Y}_n$ are regression targets sampled from the previous generation’s training targets with edits from the prior generations synthetic data:

$$\tilde{Y}_n^\top = M_{n-1} \hat{Y}_n + (1 - M_{n-1}) \tilde{Y}_{n-1}$$

Here, M_k is a diagonal matrix of 1’s or 0’s indicating whether to replace or not replace a label with a synthetic value. While Zhu et al. (2024) claim that this prevents model collapse, the proof of their test error bound (Theorem 2) assumes that $\|M_i\| = \eta \|M_{i-1}\|$ for some constant $\eta \in (0, 1)$, meaning that the number of edits decreases by at least some fixed proportion each generation. This geometric decay guarantees that the total number of edits at any given generation is finite, and their proof fails without this. However, training GPT-2 on the Natural-Instructions dataset (Mishra et al., 2021) yields only a slight decline in edit percentage (Zhu et al., 2024). Moreover, if the number of edits is finite, each generation trains on mostly real data.

4.4. Timescales of Collapse

Another key dimension of consideration is the timescale of model deterioration, which is often omitted. For example, Shumailov et al. (2023) introduced a simple theoretical setting for studying model collapse: a discrete distribution (picture a histogram) with N outcomes (atoms):

$$p^{(0)}(x) = \sum_{n=1}^N w_n \delta_{x_n}(x), \quad (1)$$

where $\sum_n w_n = 1$. If one sequentially draws D data from this distribution and computes a new distribution based on the empirical proportions, then this process forms a Markov chain with N absorbing states, each corresponding to a totally collapsed distribution, i.e., a distribution comprised of exactly one outcome. Consequently, one can use standard results from absorbing Markov chains to show that this simple process *must* collapse. However, one can go beyond a guarantee of collapse and ask: *how many model-fitting iterations can we survive before total collapse consumes us?* Again using standard results (Brydges, 2009), the expected number of model-fitting iterations before total collapse is:

$$\mathbb{E}[\text{Model Iterations Till Total Collapse}] \propto D H[p^{(0)}],$$

where $H[\cdot]$ is the Shannon entropy. If our starting distribution is uniform, the entropy is $\log(N)$ and the expected number of model-fitting iterations is $D \log(N)$. We confirmed this using numerical simulations starting from uniform distributions (Fig. 4 Left). We also numerically simulated how

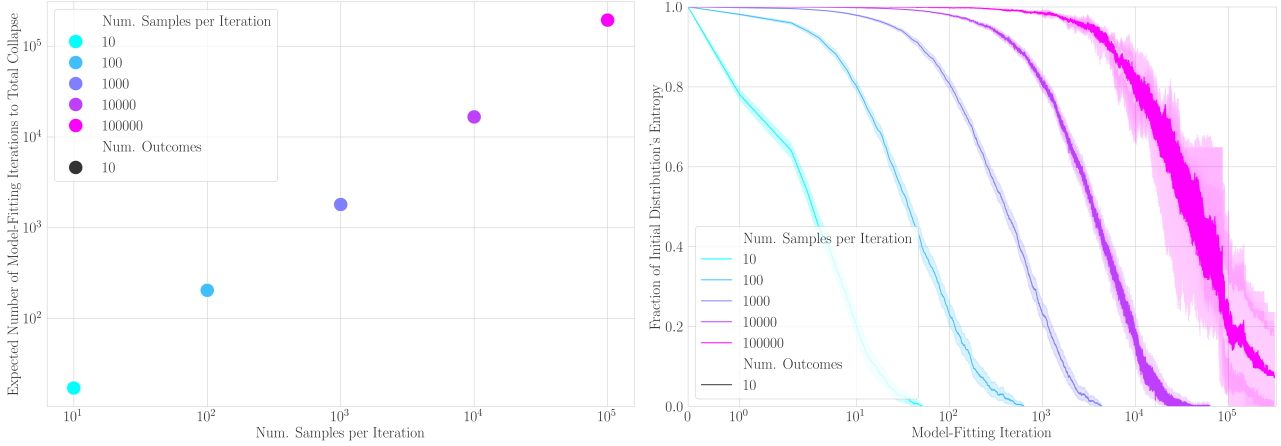


Figure 4. Dimension of Consideration for Model-Data Feedback Loops: Timescales of Collapse. Characterizing the timescale over which one should expect collapse is an underappreciated but crucial consideration. Focusing on the discrete model of Shumailov et al. (2023), the expected number of model-fitting iterations before total collapse is proportional to the number of data times the entropy of the initial data distribution (left). Taking this model at face value, this means that an enormous number of models can be trained before glimpsing the onset of collapse. However, total collapse is only the most extreme outcome; in this model, we additionally show how the entropy of the initial data distribution decays over time (right), so partial degradation such as loss of tail information can begin long before total collapse. Error bars are over 100 seeds (0 to 99, inclusive); for details, see Sec. 4.4.

quickly the entropy of $\hat{p}^{(t)}$ falls relative to the entropy of $p^{(0)}(x)$ to provide a description of the process more nuanced than just total collapse (Fig. 4 Right), since one might also be interested in how quickly tail information is lost.

While this mathematical model is simple, if we take this model at face value, we realize total model collapse poses virtually no present threat. This is because the number of real public text data alone is on the order a quadrillion tokens (Villalobos et al., 2024), and text, image, video and agentic data are high dimensional with large entropy, meaning total collapse occurs so imperceptibly slowly that humanity could train trillions of models before noticing the onset of collapse. However, this highlights a more general point: characterizing the timescale over which one should expect model collapse to occur is an underappreciated but crucial consideration for describing the model collapse profile.

Several authors do characterize timescales. Suresh et al. (2024) give an exact formulation of model collapse rate in the fundamental setting of recursive maximum likelihood estimation for discrete distributions and Gaussian mixtures; they find that for $\text{Bern}(\mu)$, $\text{Pois}(\lambda)$, and Gaussian mixtures with shared variance σ^2 distributions, the parameters μ , λ , and σ collapse to 0 exponentially in the number of generations. While Seddik et al. (2024) previously controlled the total collapse probability in both the fully synthetic and partially-synthetic recursive training regimes, Suresh et al. (2024) further control the number of unique symbols after k generations in the fully synthetic regime. Lastly, Kazdan et al. (2024b) note, in the context of kernel density estimators with fixed bandwidths, that the negative log likelihood does diverge asymptotically, although “this occurs at a rate

so glacial that it doesn’t pose a practical concern.”

5. Position: The Field Should Shift Focus to Real World Threats from Model Collapse

In conclusion, is model collapse a threat? In our view, model collapse is a multifaceted phenomenon, and a single answer is not possible. By giving greater weight to results from papers whose assumptions match real-world conditions (Section 3) and less weight to results from papers assuming conditions we argue are unrealistic, we synthesize a structured qualitative forecast of model collapse under the different definitions:

Population risk will not increase catastrophically or diverge asymptotically. Given the increase in pretraining dataset size and quality, we argue that models training on accumulating synthetic data alongside real data will not suffer from catastrophic population risk increase or diverging population risk. The jury remains out on whether the proportion of real data will approach zero.

Real tail data and modes will be lost, but how many and how quickly is unclear. Loss of diversity is a real issue, with disproportionate harms oftentimes born by subgroups. It is unclear how much of the tail we will lose or which of the real data modes will become entangled, and how synthetic data affect such changes that already occur naturally. We strongly encourage more research regarding coverage and mode collapse prevention strategies for realistic settings, building on prior work such as Hashimoto et al. (2018); Ensign et al. (2018); Taori & Hashimoto (2023).

Scaling laws may change with the introduction of synthetic data. The precise nature of these changes under realistic conditions remains to be determined. Synthetic data could potentially remove what some researchers describe as a data bottleneck, but at the cost of altered scaling. We encourage further research into how synthetic data affects scaling to better characterize likely future outcomes.

We emphasize that while real threats do exist, the popular perception that synthetic data on the internet will render future frontier AI models pretrained on web-scale data useless is likely unrealistic since such failures appear in conditions that do not match what is done in practice. Subtle degradations in data distributions might still insidiously occur, such as loss of real tail data, and future work should aim to explore what can be used to counter such outcomes.

The most likely real-world threats are quiet, not catastrophic. The threats most likely to materialize differ from those most often discussed precisely in their visibility: rather than an obvious collapse of frontier models, they are subtle degradations that pass through standard evaluation undetected. We give two concrete examples. First, aggregate metrics such as perplexity or FID can remain stable while coverage of the long tail erodes; a language model’s perplexity may look healthy even as its ability to handle minority dialects or rare medical terminology degrades. Second, synthetic data may alter the relationship between model performance and dataset size, so scaling curves calibrated on purely real data will mispredict performance on mixed real-and-synthetic corpora. Such miscalibration has direct consequences for compute budgeting at frontier labs that invest billions based on scaling projections.

5.1. Recommendations

We close with concrete recommendations, organized by audience.

For researchers. We recommend that papers report a *model collapse profile* stating explicitly which of the definitions in Sec. 2 a given result concerns, so that claims are not conflated across incompatible notions of collapse. We further recommend moving evaluation beyond population risk to include coverage, diversity, and tail-sensitive metrics, and prioritizing settings in which synthetic data accumulate alongside real data rather than replace it, as accumulation better matches current practice (Sec. 3).

For practitioners and frontier labs. We recommend investing in techniques that preserve tail coverage and diversity, and sequestering high-quality real data so it remains available for future training rather than being diluted or lost as synthetic data proliferate.

6. Alternative Views

One may feel the conditions identified in Sec. 3 are inaccurate, disproportionately emphasized, likely to change, or ignorant of important settings other than pre-training. One might also believe that the identified model collapse definitions in Sec. 2 do not fully encompass the literature or are inaccurately applied in Fig. 2. Finally, a concerned bystander could argue that the benefits of being over-cautious outweigh the costs: if society fails to anticipate model collapse by allowing wanton generation of poor-quality synthetic data, then the internet could become flooded with low-quality samples that preclude future progress. These are valid points, and we look forward to engaging with researchers and policymakers to better identify what matters to them and what the future looks like in those directions.

Impact Statement

This position paper clarifies the scientific discourse surrounding model collapse, a phenomenon often sensationalized as an unavoidable catastrophe for future generative AI. By identifying eight distinct definitions of collapse and highlighting the unrealistic assumptions—such as the total deletion of real-world data—prevalent in existing literature, we provide a framework for more rigorous and practically relevant research. Our work aims to prevent inappropriate technical responses and policy decisions that may arise from a misunderstood narrative of “poisonous” synthetic data.

Furthermore, we redirect research attention toward the more immediate threat of diversity collapse. While the most catastrophic scenarios appear readily avoidable through standard industry practices, the insidious loss of real tail data and the entanglement of data modes remain genuine risks. These degradations can disproportionately impact marginalized groups and historically disadvantaged communities, making the preservation of data diversity a critical priority for the fair development of frontier AI systems.

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A. A Case Study of Disagreement Between Papers

The wide variety of non-equivalent definitions for model collapse often creates apparent contradictions between papers claiming to study the same phenomenon. A representative example presents itself in the study of model collapse for linear regression. In this data setting, one begins with a dataset (X, y) where we assume that

$$y \sim X\beta + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma \cdot I).$$

In the first iteration, one computes

$$\hat{\beta}^{(1)} = (X^T X)^{-1} X^T y,$$

and uses the fit parameter to generate new data $(X, y^{(1)})$ with

$$y^{(1)} = X\hat{\beta}^{(1)} + \epsilon_1, \epsilon_1 \sim \mathcal{N}(0, \sigma \cdot I).$$

One can then fit successive model iterations using an *accumulate* paradigm, in which the data for the n th model fitting takes the form

$$\left(\begin{bmatrix} y, y^{(1)}, y^{(2)}, \dots, y^{(n-1)} \end{bmatrix}^T, \begin{bmatrix} X, X, \dots, X \end{bmatrix}^T \right)$$

One can also fit the n th model iteration using a *replace* paradigm, in which $\hat{\beta}^{(n)}$ is computed using only the data $(X, y^{(n-1)})$. As proven by Gerstgrasser et al. (2024), in the accumulate paradigm, the ratio

$$\frac{\mathbb{E} \left[\|\hat{\beta}^{(n)} X_{\text{test}} - y_{\text{test}}\|^2 \right]}{\mathbb{E} \left[\|\hat{\beta}^{(1)} X_{\text{test}} - y_{\text{test}}\|^2 \right]}$$

monotonically increases before converging to $\pi^2/6$. Citing Definitions 3 and 4, Gerstgrasser et al. (2024) correctly asserted that model collapse does not occur. However, if one instead defines model collapse by Definition 2, this scenario does exhibit collapse.

To complicate the story, Dohmatob et al. (2024a) studied the same model under the replace paradigm. Under the replace paradigm, Definition 3 suggests that model collapse occurs since the asymptotic risk diverges, while Definition 4 implies that model collapse does not occur, since the replace scenario does not exhibit vanishing variance. Table 1 shows how incompatible definitions lead to confusion.

Table 1. Model collapse occurs under some definitions, but does not occur under others for the regression setting described in Appendix A.

Definition	Accumulate	Replace
Def. 1	✗	✓
Def. 2	✓	✓
Def. 3	✗	✓
Def. 4	✗	✗
Def. 5	✓	✓
Def. 6	✗	✗
Def. 7	✗	✗
Def. 8	✗	✗